10.5 The Riemann–Lebesgue Lemma

The Riemann–Lebesgue lemma is a basic result of Fourier analysis. In this course it is what lets us drop the multi-particle and vacuum pieces of the asymptotic limits in the LSZ derivation (§10.6), and it is behind the common phrase "the integrand oscillates rapidly and averages to zero".

Statement of the Lemma

Let $f \in L^1(\mathbb{R}^n)$ (Lebesgue integrable). The Fourier transform of f,

$$\mathcal{F}[f](\xi) = \int_{\mathbb{R}^n} f(x)\, e^{-i\xi\cdot x}\, dx,$$

satisfies

$$\lim_{|\xi|\to\infty} \mathcal{F}[f](\xi) = 0.$$

The same holds for the inverse transform of an L^1 function.

Key Intuition, and a Common Confusion

The lemma reflects destructive interference from rapid oscillations in $e^{-i\xi\cdot x}$ at large $|\xi|$. This occurs universally for L^1-functions, regardless of their smoothness or decay.

Critical point: the vanishing applies to the *transform*, not to the original function. If, on top of being Lebesgue integrable, $f(x)$ is smooth with compact support (the set where $f \neq 0$ is bounded), then $f(x)$ vanishes at infinity as well, but for a different reason. When people show examples, cases with and without compact support are often mixed, which suggests that the Riemann–Lebesgue lemma also implies a vanishing $f(x)$ at infinity. It does not!

Lebesgue Integrability

A function f is **Lebesgue integrable** ($f \in L^1$) if

$$\int_{\mathbb{R}^n} |f(x)|\, dx < \infty.$$

This asks only that the total area under $|f|$ be finite; it is not a condition on the total variation of f. Key properties:

- It does *not* guarantee $f(x) \to 0$ as $|x| \to \infty$.
- It allows functions with non-compact support (e.g., $f(x) = e^{-|x|}$).
- It is insensitive to changes on sets of measure zero.

Decay of the Original Function

- **No automatic decay**: L^1-integrability alone does not imply $f(x) \to 0$. Example:

$$f(x) = \sum_{n=1}^{\infty} n \cdot \chi_{[n,\, n+1/n^3]}(x) \in L^1(\mathbb{R}), \qquad \int |f|\, dx = \sum_n \frac{1}{n^2} < \infty,$$

  where $f(x)$ has spikes of growing height at $x = n \to \infty$, while its Fourier transform still goes to zero.
- **Sufficient conditions for decay**:
  - $f \in C_c^\infty(\mathbb{R}^n)$ (smooth and compactly supported);
  - $f \in \mathcal{S}(\mathbb{R}^n)$ (Schwartz space: rapid decay);
  - f in the Sobolev space $W^{k,p}(\mathbb{R}^n)$ with $kp > n$, $p < \infty$.

Compactly Supported Smooth Functions

For $f \in C_c^\infty(\mathbb{R}^n)$:

- **Vanishing at infinity**: trivially guaranteed by compact support.
- **Stronger than** L^1: all C_c^∞-functions are L^1, but not vice versa.
- **Rapidly decaying transform**: smoothness makes $\mathcal{F}[f]$ fall faster than any power.
- **Uses in physics**: test functions for distributions (e.g., the Dirac delta), smearing of field operators, and localized wave packets.

Examples and Counterexamples

- **Decaying L^1-function**: $f(x) = e^{-|x|}$,

$$\mathcal{F}[f](\xi) = \frac{2}{1+\xi^2} \to 0 \quad \text{as } |\xi| \to \infty.$$

- **Non-decaying L^1-function**: the spike train above, with $\mathcal{F}[f](\xi) \to 0$ but $f(x) \not\to 0$.
- **Compactly supported case**: the bump function

$$f(x) = \begin{cases} e^{-1/(1-x^2)} & |x| < 1, \\ 0 & \text{otherwise}, \end{cases}$$

  with $f(x) \to 0$ and $\mathcal{F}[f] \in \mathcal{S}(\mathbb{R})$.

Caveats

- L^1 **is needed**: the lemma fails if $f \notin L^1$ (e.g., $f(x) = 1$, whose transform is $2\pi\delta(\xi)$).
- **No decay rate**: the lemma only states $\mathcal{F}[f](\xi) \to 0$, not how fast.
- **Tempered distributions**: for $f \in \mathcal{S}'(\mathbb{R}^n)$, $\mathcal{F}[f]$ need not decay (e.g., $\mathcal{F}[\delta] = 1$).

Where It Enters QFT

- **Asymptotic limits in LSZ.** A wave-packet amplitude $\int d^3p\, w(\mathbf{p})\, e^{iK(\mathbf{p})t}$ with $w \in L^1$ is the Fourier transform of the distribution of K. If K takes a continuum of values, it goes to zero as $t \to \pm\infty$; if K takes one value, it does not. This is exactly how the multi-particle part of $a_1^\dagger(t)|0\rangle$ drops out while the one-particle part survives (§10.6).
- **"Boundary pieces do not contribute to the spectrum."** We often argue that when we Fourier transform fields at spacetime infinity, the integrand oscillates rapidly and destructively interferes, so the boundary piece does not contribute. Read correctly, this is the lemma applied to a smooth, integrable *envelope* (a wave packet, a smeared field), not to a plane wave.
- **Fields are distributions.** Quantum fields are operator-valued tempered distributions. A free field $\varphi(x)$ does not vanish at spatial infinity, and its Fourier transform is concentrated on the mass shell rather than decaying. Physical statements are made after smearing with test functions, $\varphi(f) = \int d^4x\, f(x)\varphi(x)$, and it is the smooth f that brings decay.
- **Hilbert spaces.** In quantum mechanics, $L^2(\mathbb{R}^3)$ (Lebesgue square-integrable wavefunctions) is the Hilbert space; the Lebesgue integral is what makes it complete.

Summary

- The Riemann–Lebesgue lemma ensures that Fourier transforms of L^1-functions vanish at infinity.
- L^1-integrability does *not* imply $f(x) \to 0$; compact support or rapid decay are required for that.
- Compactly supported smooth functions are a strict subset of L^1, and they are the test functions and wave packets of QFT.